# CSE 311:
# Data Communication

**Instructor:**

**Dr. Md. Monirul Islam**

# Signals and systems

## Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

# Fourier's theorem for aperiodic Signal

Revisited!

- *aperiodic* signal but *with finite area under curve* can be represented as *integral* of sines and cosines

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j\omega t} df$$

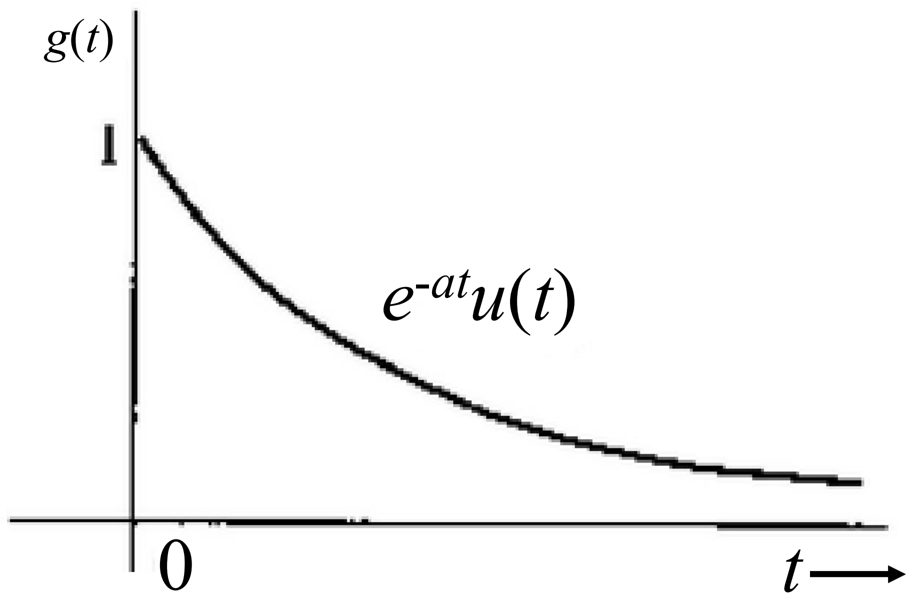
$$\text{where, } G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$\text{and } \omega = 2\pi f$$

# Fourier Transform Example

Revisited!

What happens if result for $a < 0$?




$$G(f) = \frac{-1}{a + j2\pi f}\left[e^{-(a+j2\pi f)t}\right]_0^\infty$$

$$= \frac{1}{a + j2\pi f}$$

Now as $t$->$\alpha$, $e^{-(a+j2\pi f)t} = e^{-at}\, e^{-j2\pi f)t} = 0$ provided $a > 0$

# Existence of Fourier Transform

Revisited!

Conditions

- Area under curve must be finite
- The function must be integrable

$$\int_{-\infty}^{\infty} |g(t)|\, dt \ < \ \infty$$

Dirichlet condition

# Existence of Fourier Transform

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- The function must be integrable

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This can be proved from,

$$|G(f)| = \left| \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt \right|$$

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$$|G(f)| \le \int_{-\infty}^{\infty} |g(t)|\, dt$$

# Linearity of Fourier Transform

Assume, $g_1(t) \Leftrightarrow G_1(f)$ and $g_2(t) \Leftrightarrow G_2(f)$

and $a_1$ and $a_2$ are arbitrary constants

Then,

$$a_1 g_1(t) + a_2 g_2(t) \Leftrightarrow a_1 G_1(f) + a_2 G_2(f)$$

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Then,

$$a_1 g_1(t) + a_2 g_2(t) \Leftrightarrow a_1 G_1(f) + a_2 G_2(f)$$

This means,

$$\Im\left[a_1 g_1(t) + a_2 g_2(t)\right] = a_1 G_1(f) + a_2 G_2(f)$$

# Linearity of Fourier Transform

More general case

$$\sum_k a_k g_k(t) \Leftrightarrow \sum_k a_k G_k(f)$$

# Linearity of Fourier Transform

More general case

$$\sum_k a_k g_k(t) \Leftrightarrow \sum_k a_k G_k(f)$$

Superposition Theorem

# Area Under Curve of Fourier Transform *G(f)*

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

# Area Under Curve of Fourier Transform *G*(*f*)

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

Assume $t = 0$

$$g(0) = \int_{-\infty}^{\infty} G(f) e^{j2\pi f.0} df = \int_{-\infty}^{\infty} G(f) df$$

# Area Under Curve of Time domain Function *g*(*t*)

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$$

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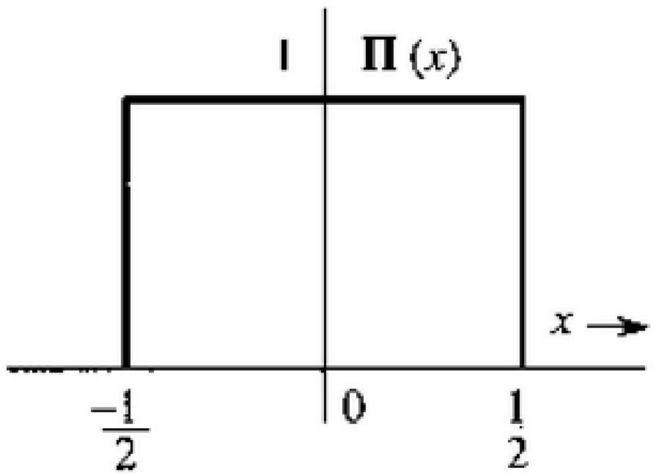
# Fourier Transform of Some Useful Functions

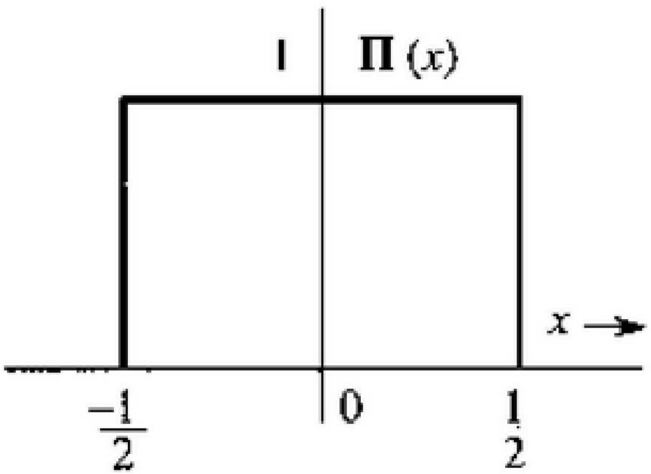
Unit Rectangular Function




$$\Pi(x) = \begin{cases} 1 & |x| \le \frac{1}{2} \\ 0.5 & |x| = \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$$

# Fourier Transform of Some Useful Functions

Unit Rectangular Function




Other names

Box Function

Gate Function

$$\Pi(x) = \begin{cases} 1 & |x| \le \frac{1}{2} \\ 0.5 & |x| = \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$$

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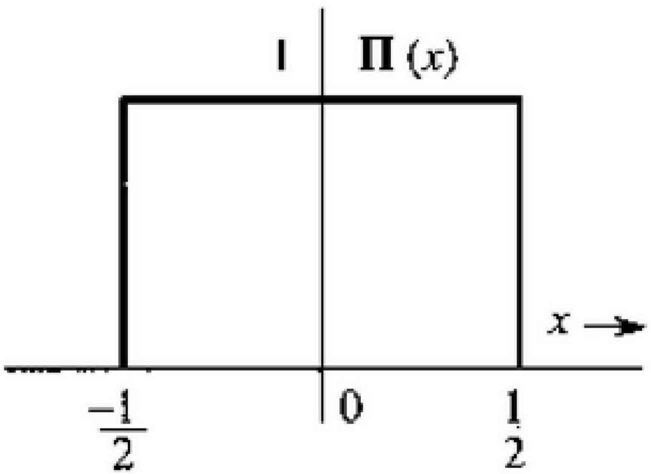
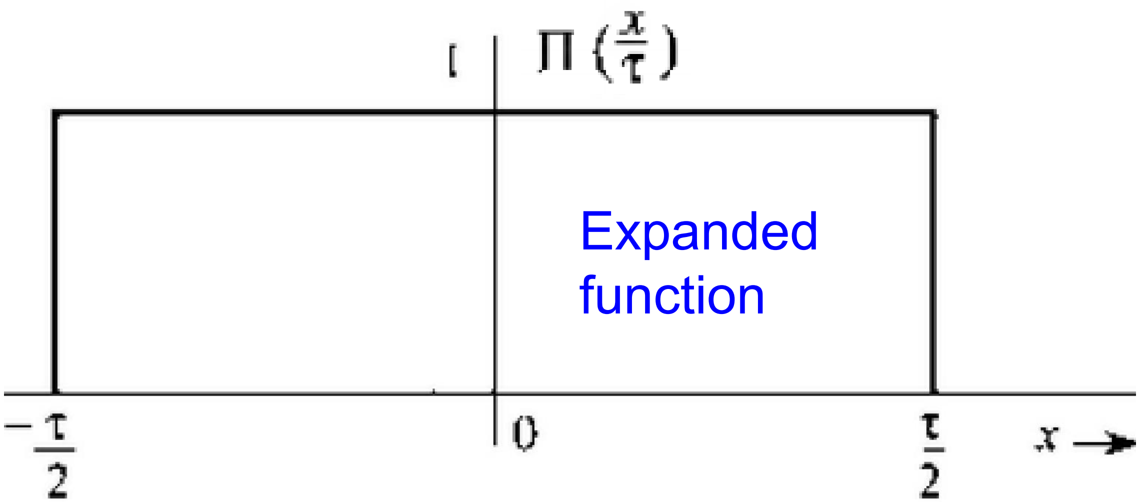


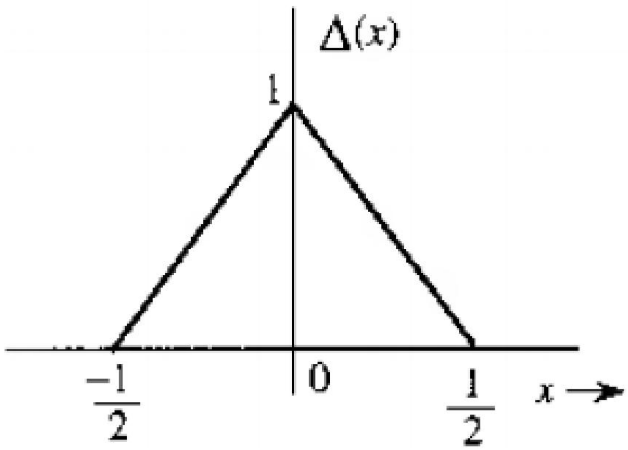


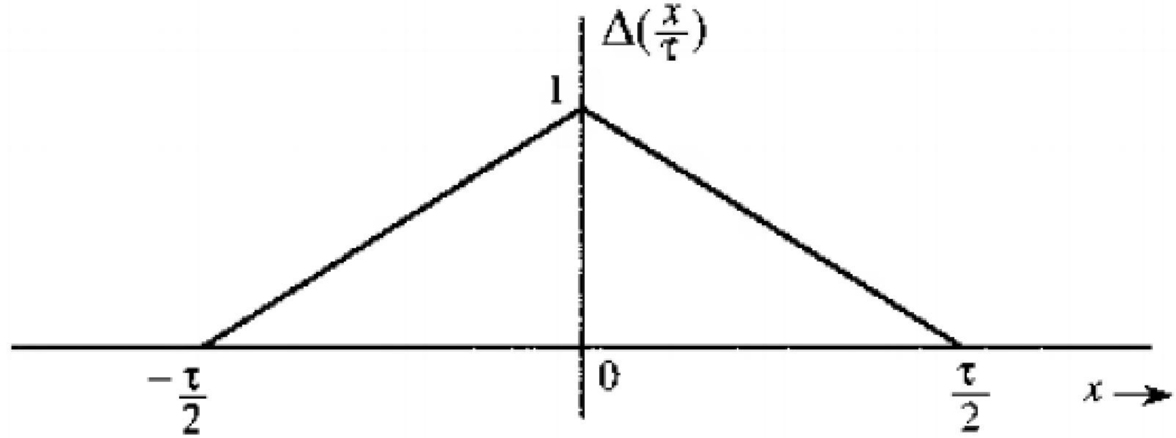

$$\Pi(x) = \begin{cases} 1 & |x| \le \frac{1}{2} \\ 0.5 & |x| = \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$$

$$\Pi\left(\frac{x}{\tau}\right) = \begin{cases} 1 & |x| \le \frac{\tau}{2} \\ 0.5 & |x| = \frac{\tau}{2} \\ 0 & |x| > \frac{\tau}{2} \end{cases}$$

# Fourier Transform of Some Useful Functions

Unit Triangular Function



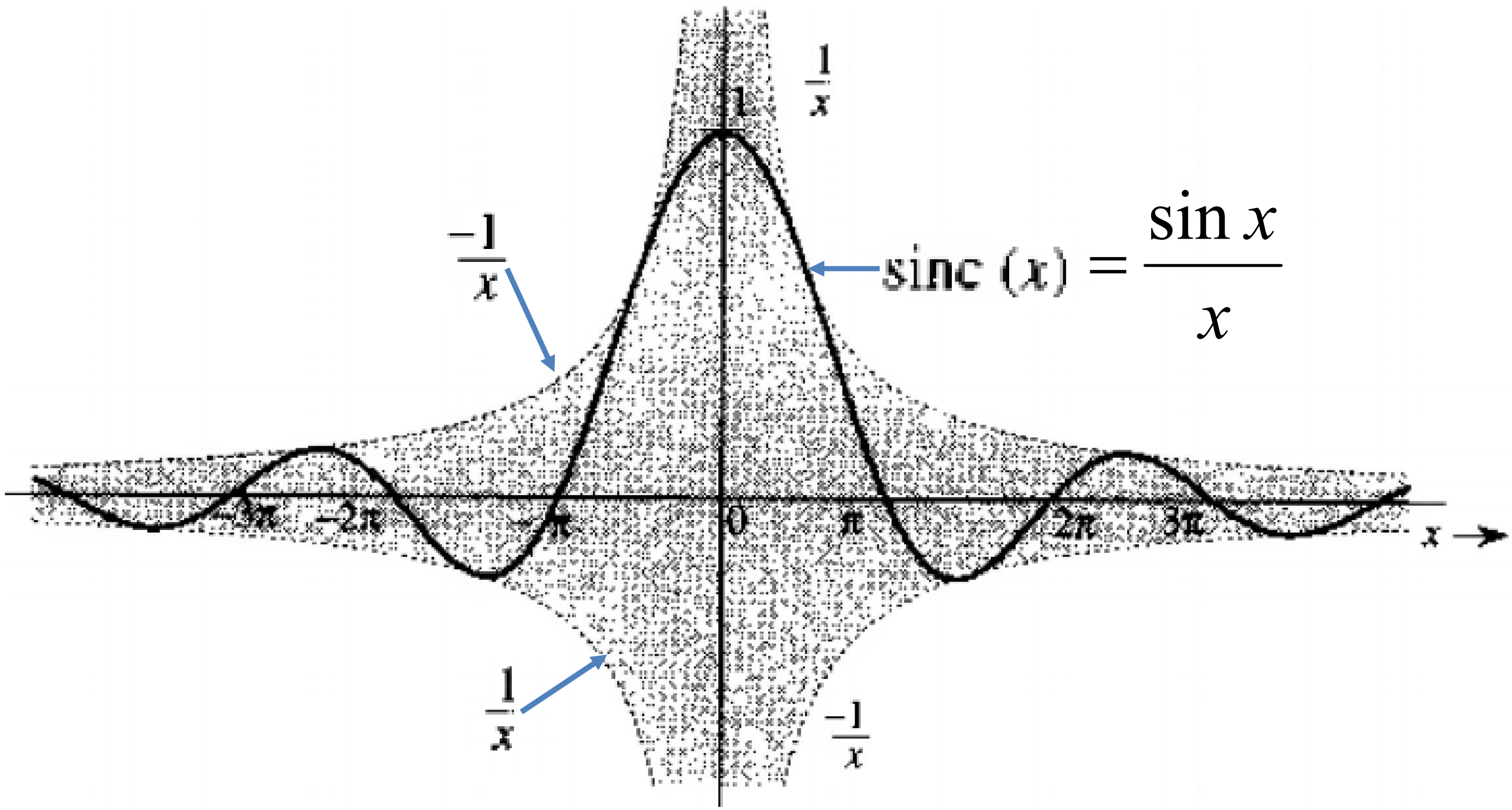




$$\Delta(x) = \begin{cases} 1 - 2|x|, & |x| < \frac{1}{2} \\ 0, & |x| > \frac{1}{2} \end{cases}$$

$$\Delta(\tfrac{x}{\tau}) = \begin{cases} 1 - 2|\frac{x}{\tau}|, & |x| < \frac{\tau}{2} \\ 0, & |x| > \frac{\tau}{2} \end{cases}$$

# Fourier Transform of Some Useful Functions

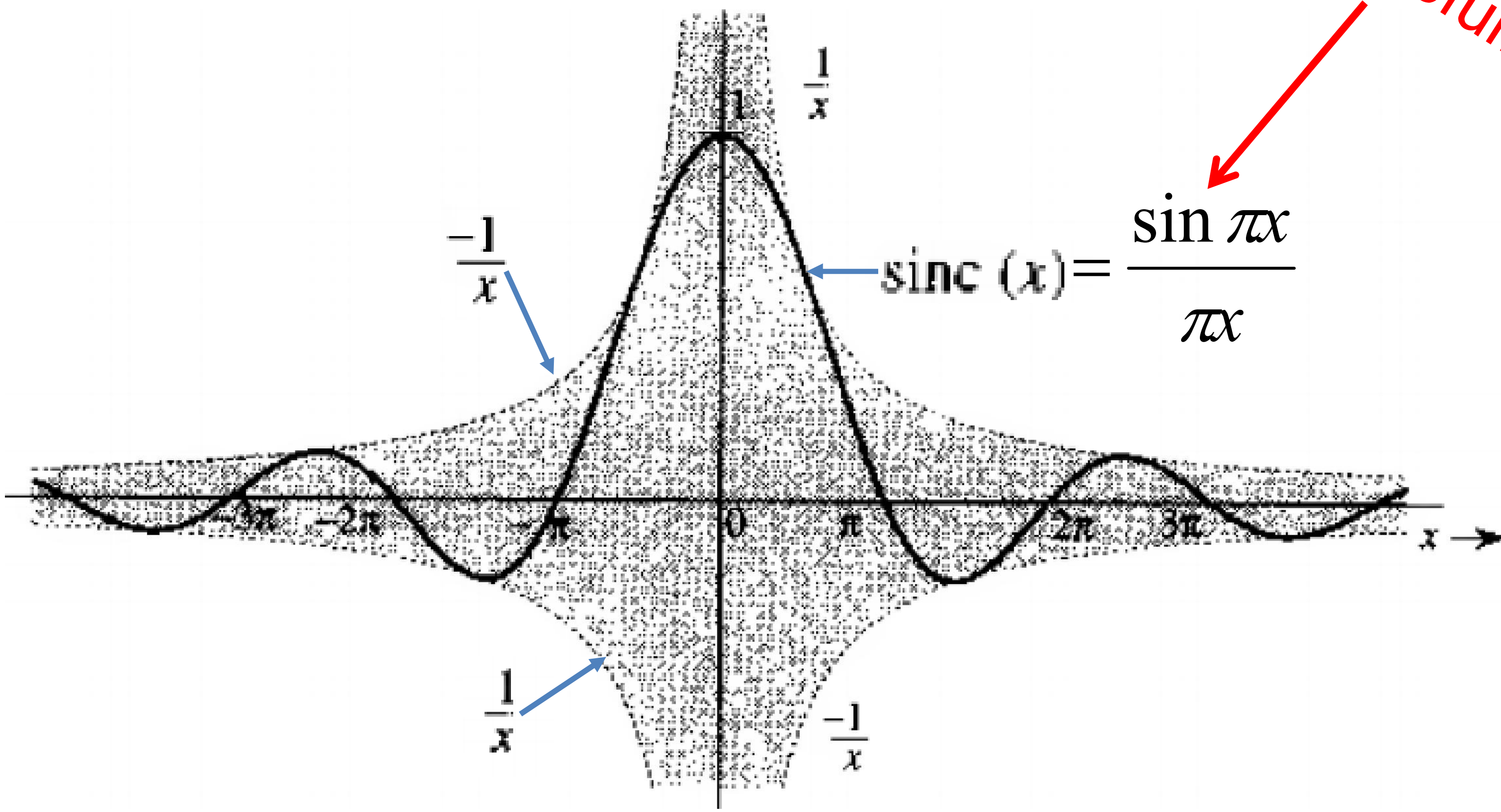
Sinc Function, $\text{sinc}\,(x)$



# Fourier Transform of Some Useful Functions

Sinc Function, $\mathrm{sinc}\,(x)$

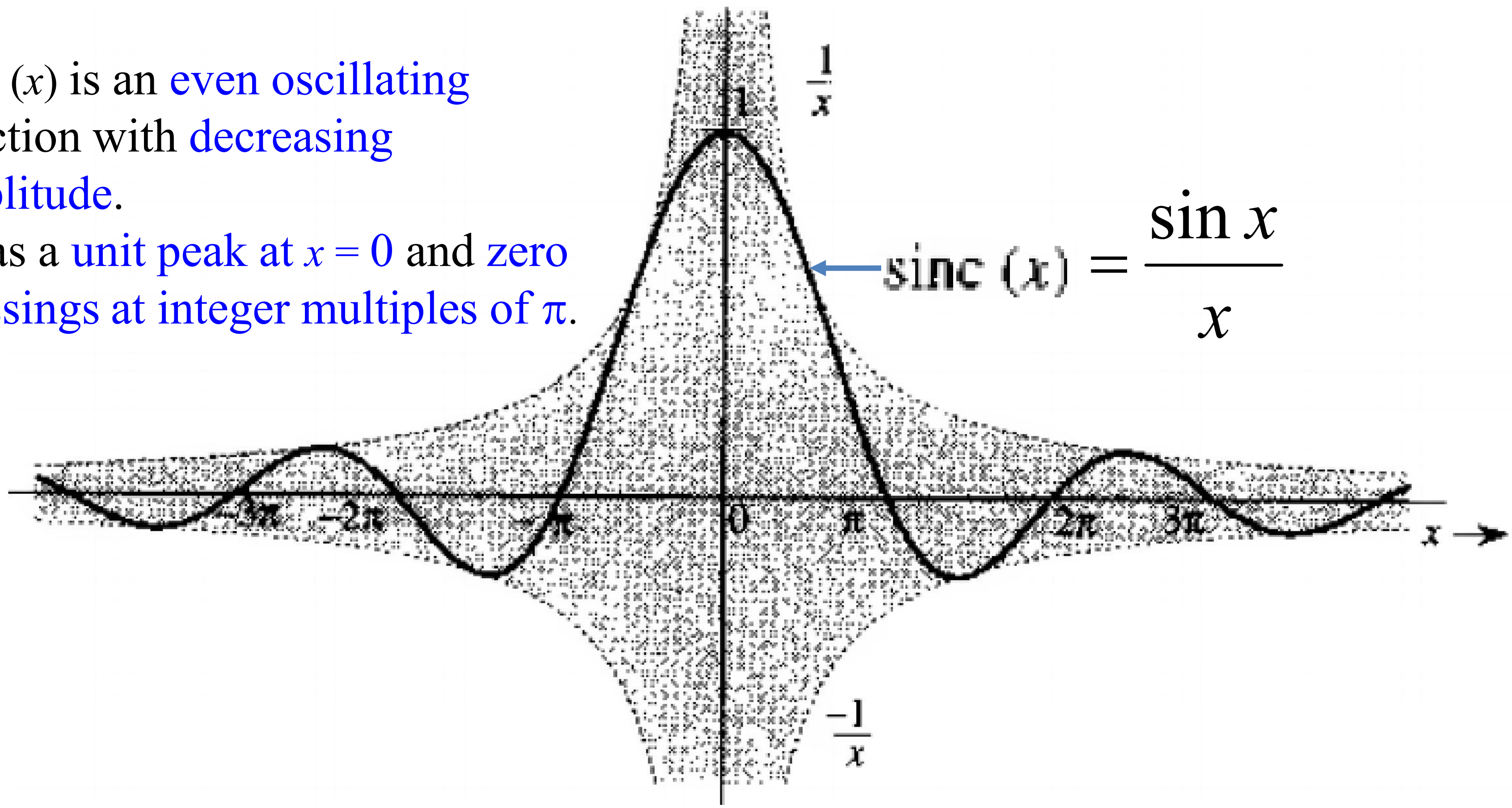
Be careful!



# Fourier Transform of Some Useful Functions

Sinc Function, sinc $(x)$

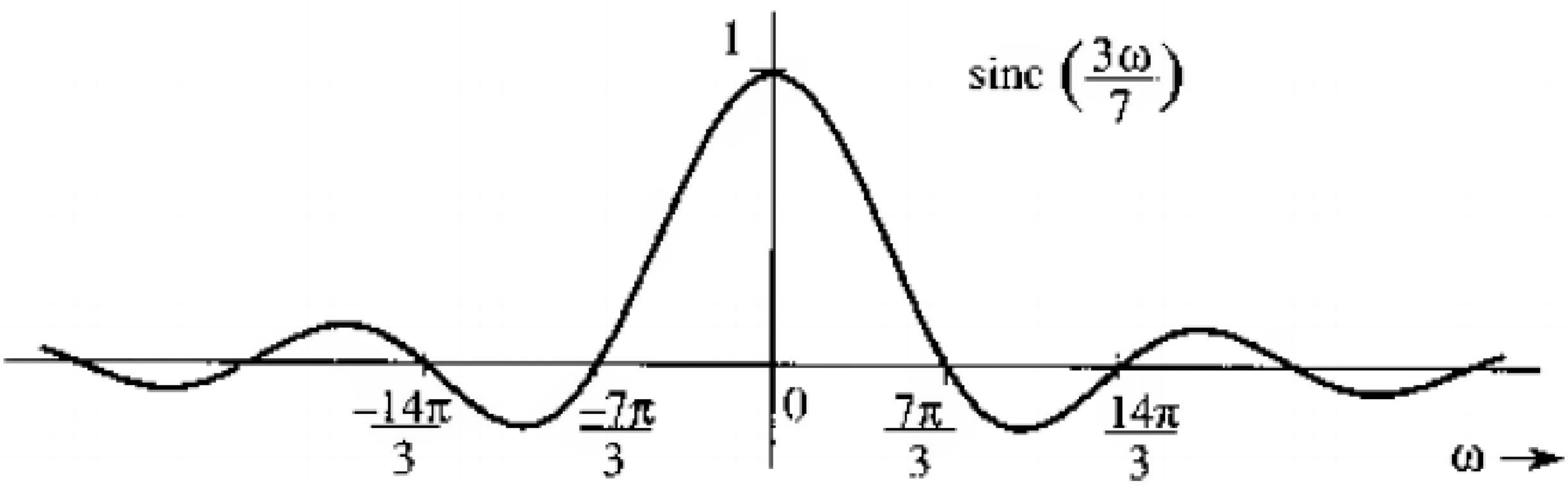
- sinc $(x)$ is an even oscillating function with decreasing amplitude.
- It has a unit peak at $x = 0$ and zero crossings at integer multiples of $\pi$.




$$\text{sinc}(x) = \frac{\sin x}{x}$$

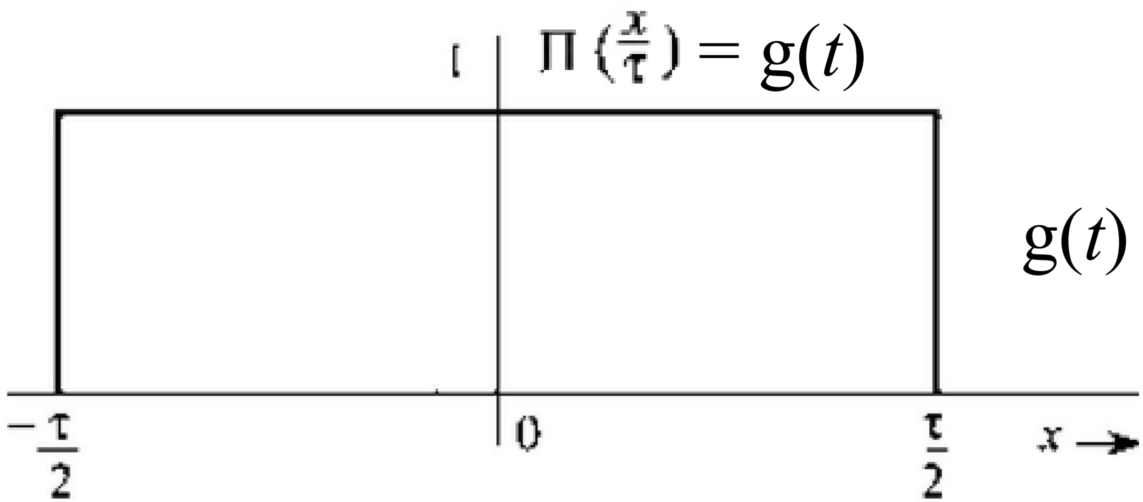
# Fourier Transform of Some Useful Functions

Sinc Function, $\text{sinc}\,(x)$



# Fourier Transform of Some Useful Functions

Unit scaled rectangular function




$$g(t) = \Pi\left(\frac{x}{\tau}\right) = \begin{cases} 1 & |x| \leq \frac{\tau}{2} \\ 0.5 & |x| = \frac{\tau}{2} \\ 0 & |x| > \frac{\tau}{2} \end{cases}$$

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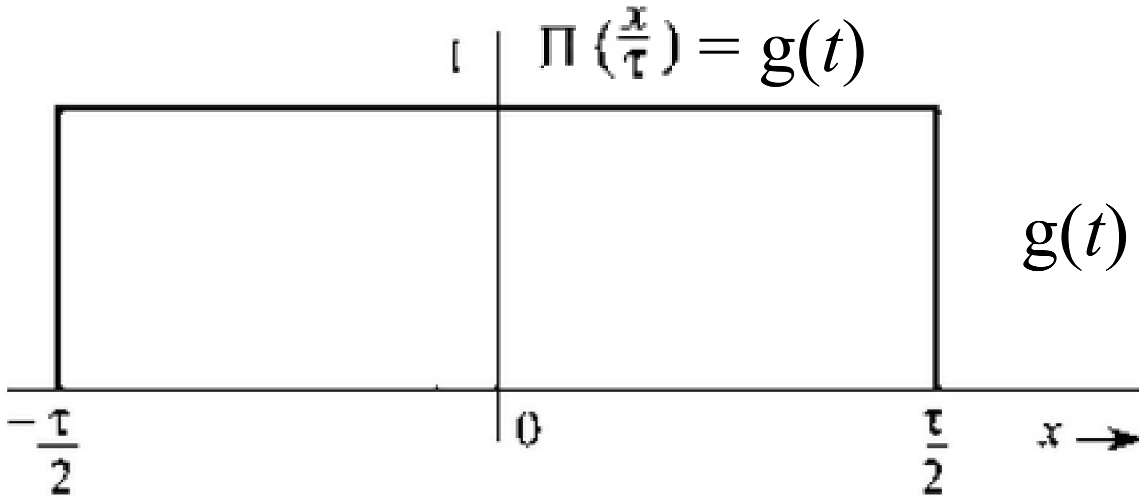


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$$G(f) = \int_{-\infty}^{\infty} \Pi\left(\frac{t}{\tau}\right) e^{-j2\pi f t}\, dt$$

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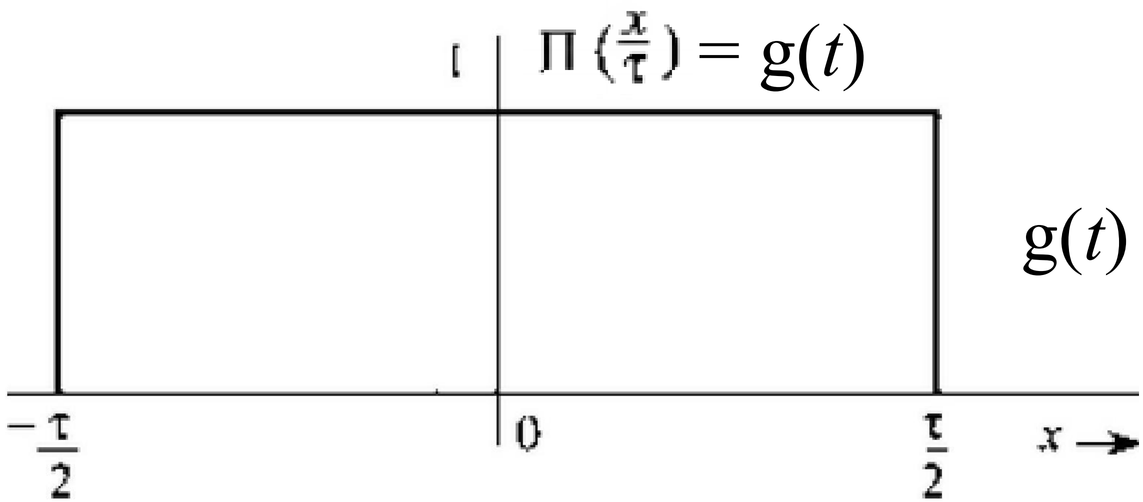


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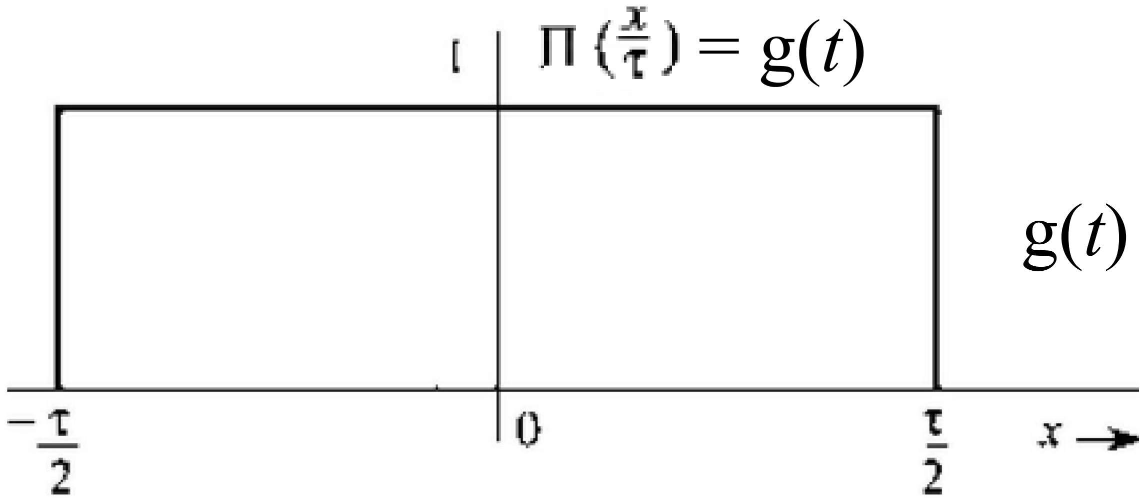


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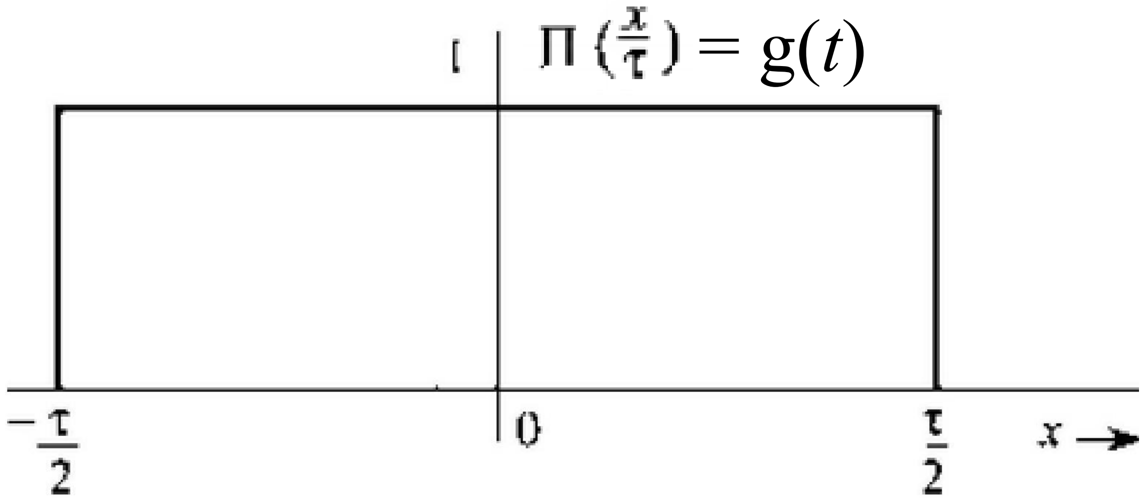
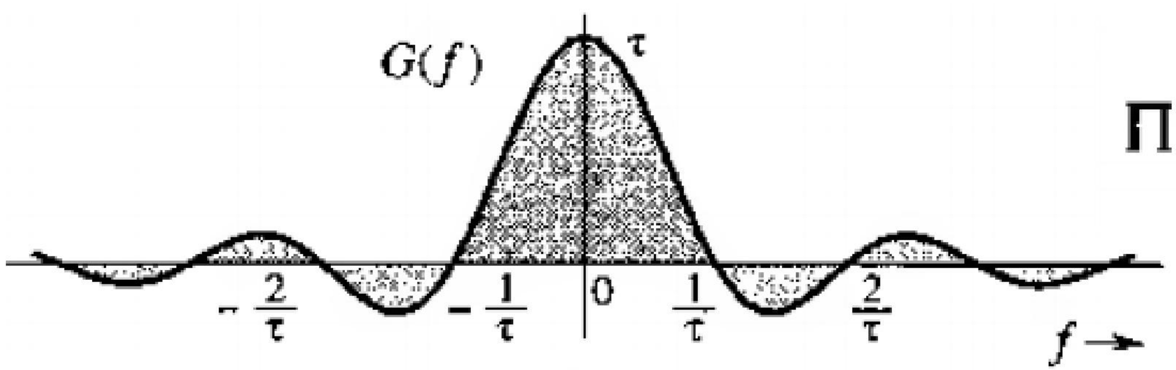
$$= -\frac{1}{j2\pi f}\left(e^{-j\pi f\tau} - e^{j\pi f\tau}\right) = \frac{2\sin(\pi f\tau)}{2\pi f} = \tau\frac{\sin(\pi f\tau)}{(\pi f\tau)} = \tau \ \mathrm{sinc}\ (\pi f\tau)$$

# Fourier Transform of Some Useful Functions

Unit scaled rectangular function



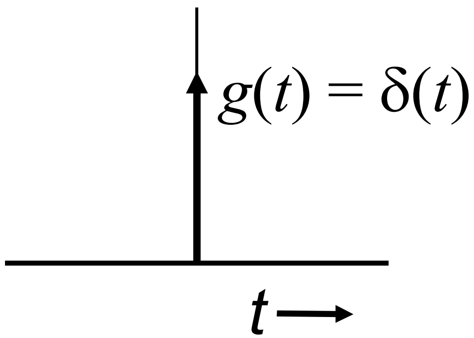




Fourier transform pair

$$\Pi\left(\frac{t}{\tau}\right) \Longleftrightarrow \tau\ \mathrm{sinc}\ (\pi f \tau)$$

# Fourier Transform of Some Useful Functions

Unit Impulse function $\delta(t)$

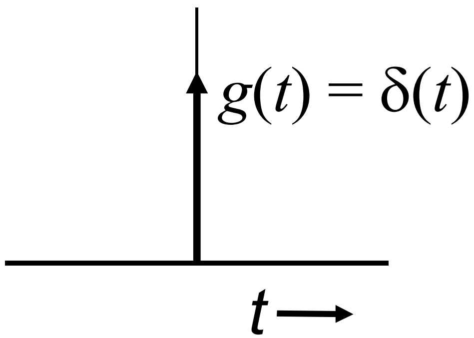


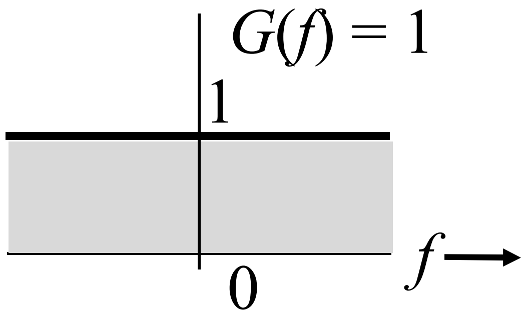

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi f t}\, dt$$

$$= e^{-j2\pi f \cdot 0} = 1$$

# Fourier Transform of Some Useful Functions

Unit Impulse function $\delta(t)$




$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi ft}\, dt$$

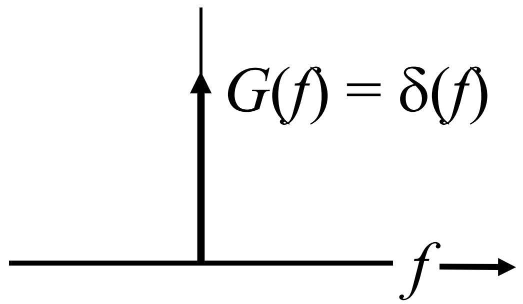
$$= e^{-j2\pi f \cdot 0} = 1$$

Fourier transform pair

$$\delta(t) \Longleftrightarrow 1$$

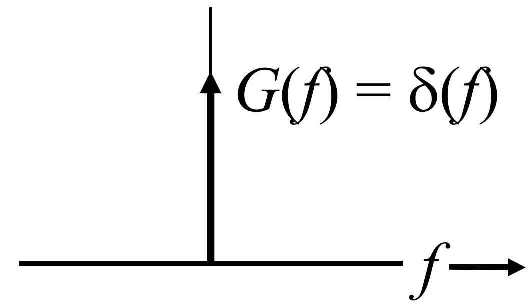
# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f)$



# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f)$



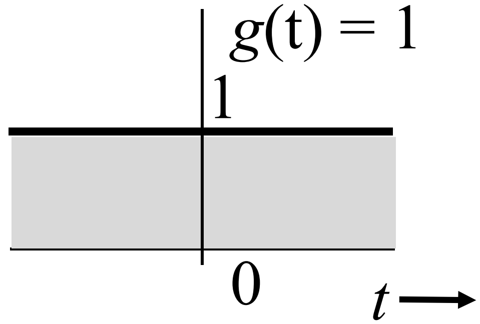

$$g(t) = \Im^{-1}[\delta(f)]$$

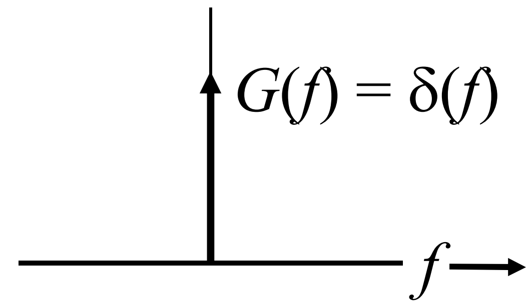
$$= \int_{-\infty}^{\infty} G(f)e^{j2\pi ft}df = \int_{-\infty}^{\infty} \delta(f)e^{j2\pi ft}df$$

$$= e^{j2\pi t.0} = 1$$

# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f)$

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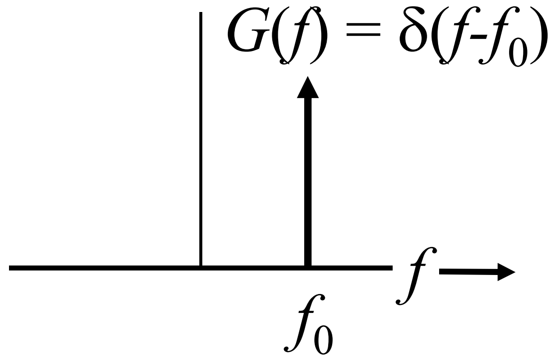
$$= e^{j2\pi t.0} = 1$$

Fourier Transform Pair

$$1 \Longleftrightarrow \delta(f)$$

# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$



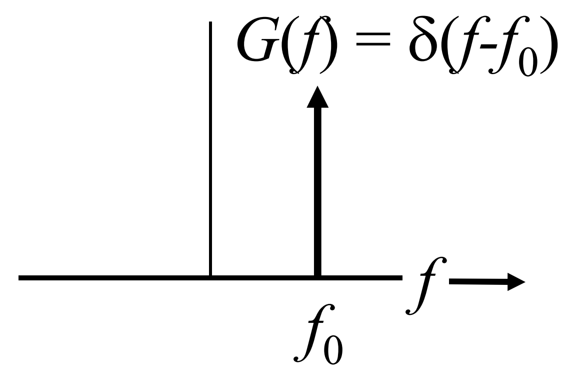
# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f\text{-}f_0)$

$$g(t) = \Im^{-1}[\delta(f - f_0)]$$

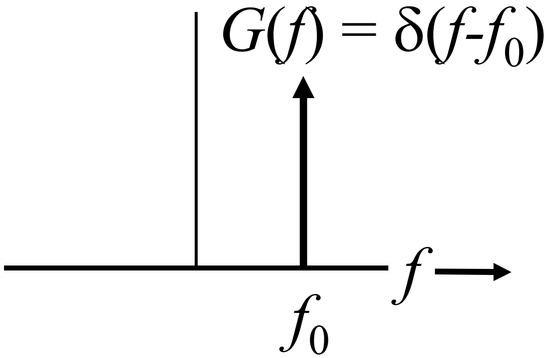
$$= \int_{-\infty}^{\infty} G(f)e^{j2\pi ft}df = \int_{-\infty}^{\infty} \delta(f - f_0)e^{j2\pi ft}df$$

$$= e^{j2\pi tf_0} = e^{j2\pi f_0 t}$$



# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$




$$g(t) = \Im^{-1}[\delta(f-f_0)]$$

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$$= e^{j2\pi tf_0} = e^{j2\pi f_0 t}$$

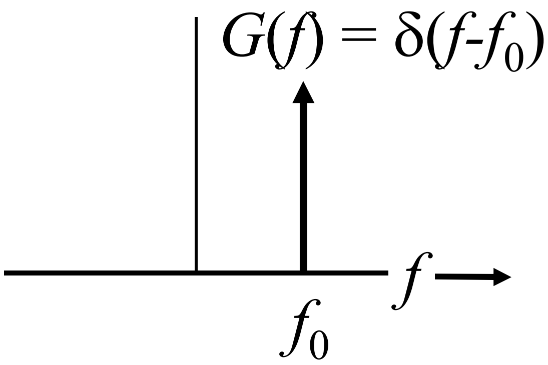
Fourier Transform Pair

$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f-f_0)$$

# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$

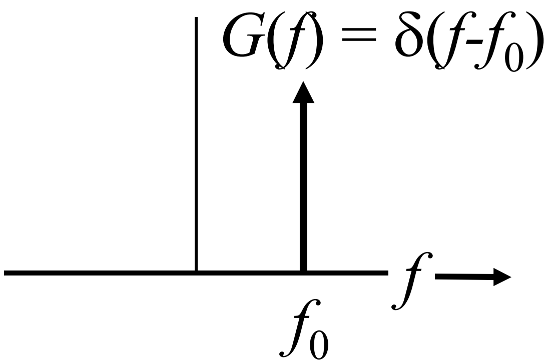
$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f - f_0)$$



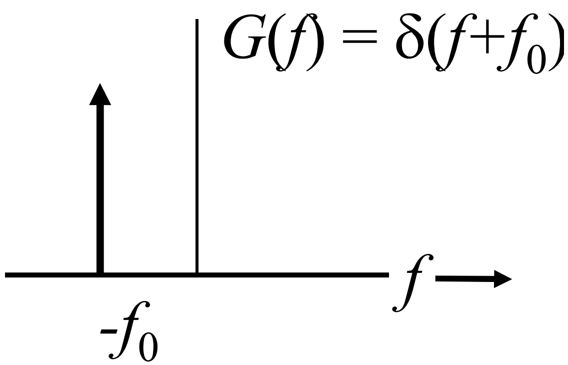
# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f-f_0)$ and $\delta(f+f_0)$

$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f - f_0)$$



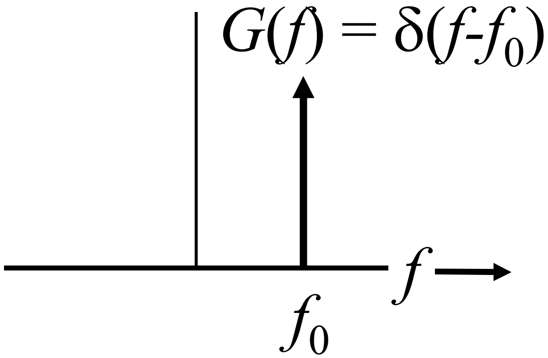

$$? \qquad \Leftrightarrow \delta(f + f_0)$$



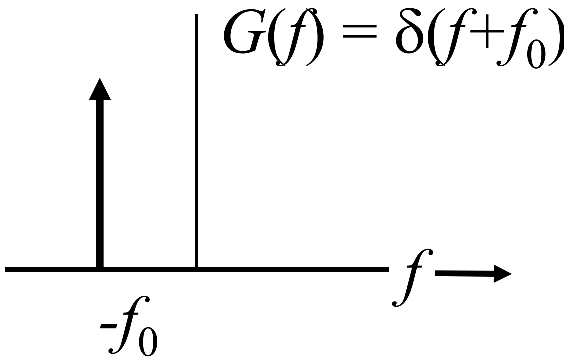
# Fourier Transform of Some Useful Functions

Inverse Fourier Transform of $\delta(f\text{-}f_0)$ and $\delta(f+f_0)$

$$e^{j2\pi f_0 t} \Leftrightarrow \delta(f - f_0)$$




$$e^{-j2\pi f_0 t} \Leftrightarrow \delta(f + f_0)$$




Fourier Transform Pairs

# Fourier Transform of Some Useful Functions

Fourier Transform of cosine function

Assuming, $e^{j\theta} = \cos\theta + j\sin\theta$

$$\cos 2\pi f_0 t = \frac{1}{2}(e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$

# Fourier Transform of Some Useful Functions

Fourier Transform of cosine function

Assuming, $e^{j\theta} = \cos\theta + j\sin\theta$

$$\cos 2\pi f_0 t = \frac{1}{2}(e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$

Fourier Transform Pair

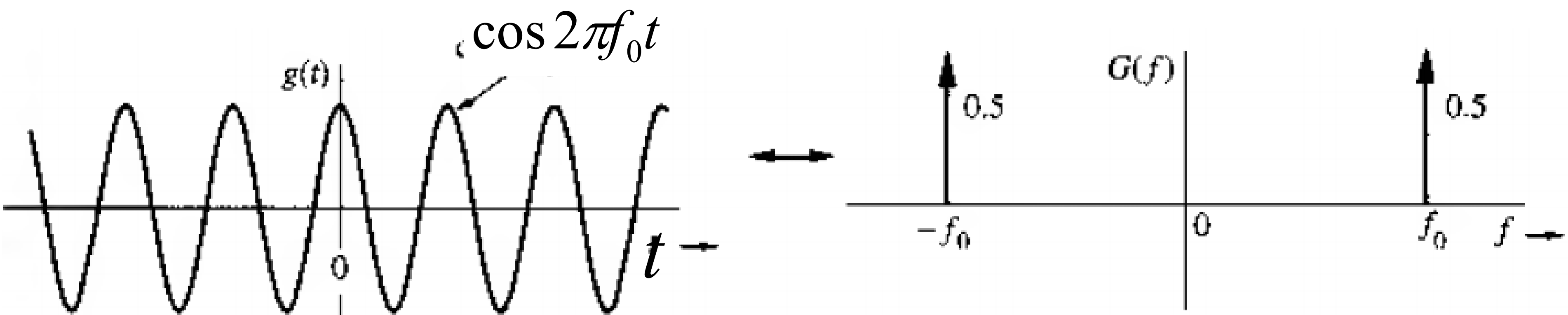
$$\cos 2\pi f_0 t \Leftrightarrow \frac{1}{2}[\delta(f - f_0) + \delta(f + f_0)]$$

# Fourier Transform of Some Useful Functions

Fourier Transform of cosine function

Assuming, $e^{j\theta} = \cos\theta + j\sin\theta$

$$\cos 2\pi f_0 t = \frac{1}{2}(e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$




$$\cos 2\pi f_0 t \Leftrightarrow \frac{1}{2}[\delta(f - f_0) + \delta(f + f_0)]$$

# Fourier Transform of Some Useful Functions

Fourier Transform of sine function

Assuming, $e^{j\theta} = \cos\theta + j\sin\theta$

$$\sin 2\pi f_0 t = \frac{1}{2j}(e^{j2\pi f_0 t} - e^{-j2\pi f_0 t})$$

# Fourier Transform of Some Useful Functions

Fourier Transform of sine function

Assuming, $e^{j\theta} = \cos\theta + j\sin\theta$

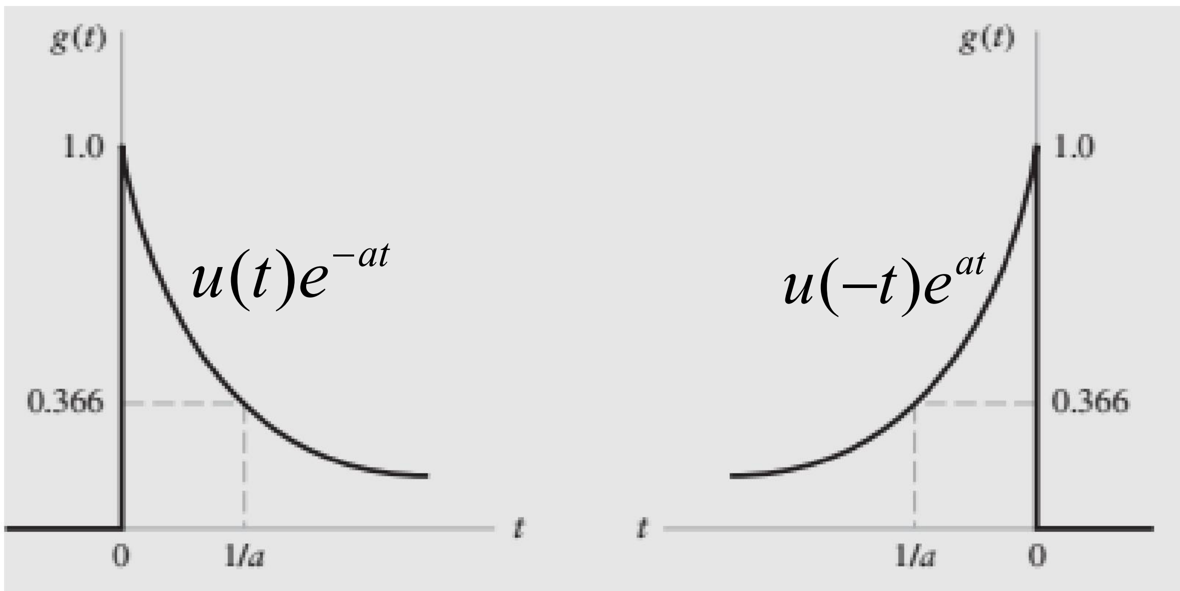
$$\sin 2\pi f_0 t = \frac{1}{2j}(e^{j2\pi f_0 t} - e^{-j2\pi f_0 t})$$

Fourier Transform Pair

$$\sin 2\pi f_0 t \Leftrightarrow \frac{1}{2j}[\delta(f - f_0) - \delta(f + f_0)]$$

# Fourier Transform of Some Useful Functions

Fourier Transform of exponential functions

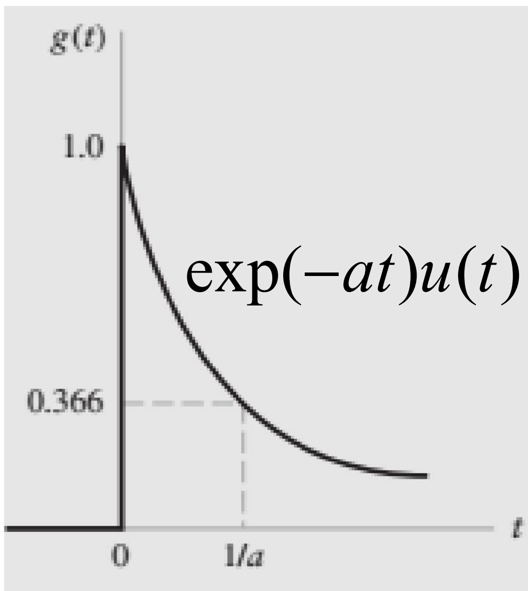



Exponential decaying fnc

Exponential rising fnc

# Fourier Transform of Some Useful Functions

## Fourier Transform of exponential functions




Exponential decaying fnc

$$G(f) = \int_0^\infty e^{-at} e^{-j2\pi f t} dt = \int_0^\infty e^{-(a+j2\pi f)t} dt$$

$$= \frac{1}{a + j2\pi f}$$

Fourier Transform Pair

$$\exp(-at)u(t) \rightleftharpoons \frac{1}{a + j2\pi f}$$

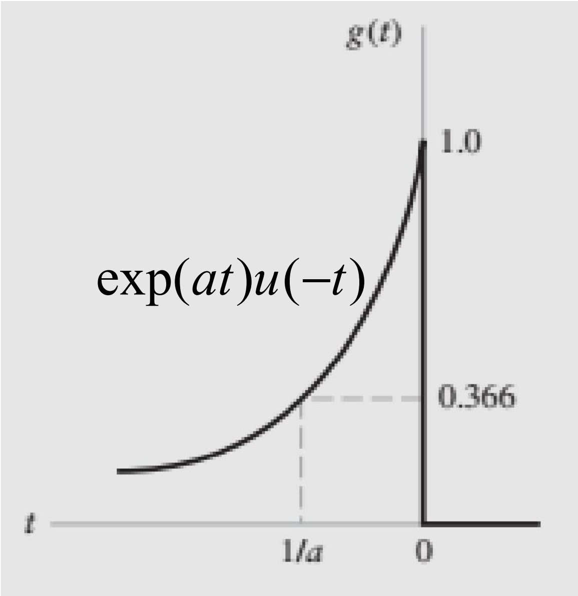
# Fourier Transform of Some Useful Functions

Fourier Transform of exponential functions

$$G(f) = \int_{-\infty}^{0} e^{at} e^{-j2\pi ft} dt$$

$$= \int_{0}^{\infty} e^{-t(a-j2\pi f)} dt$$

$$= \frac{1}{a - j2\pi f}$$



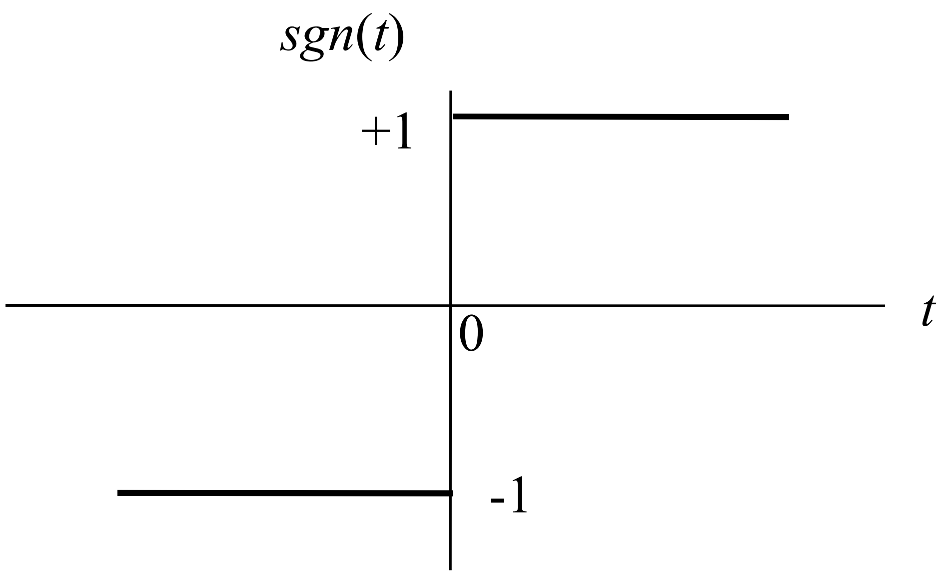

Exponential rising fnc

Fourier Transform Pair

$$\exp(at)u(-t) \rightleftharpoons \frac{1}{a - j2\pi f}$$

# Fourier Transform of Some Useful Functions

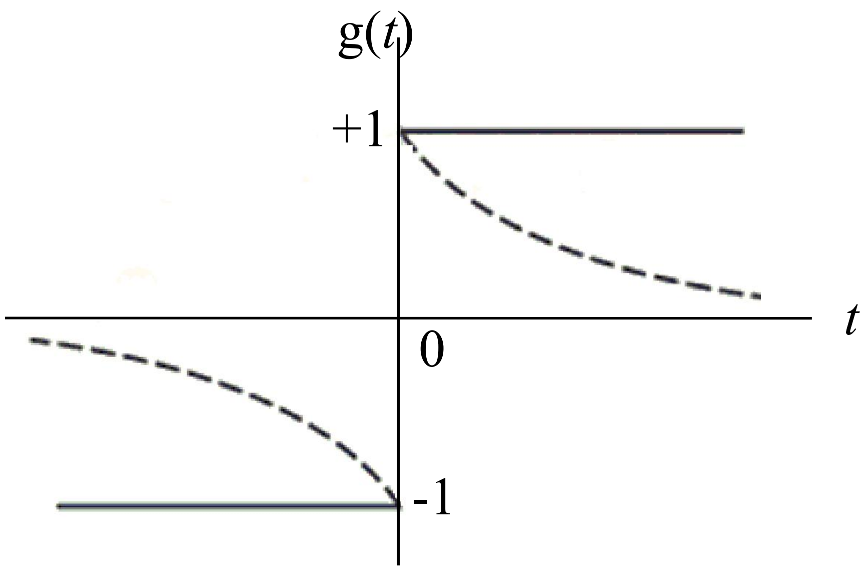
Fourier Transform of *sgn*(*t*) function




$$\operatorname{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

# Fourier Transform of Some Useful Functions

Fourier Transform of sgn($t$) function




$$\mathrm{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

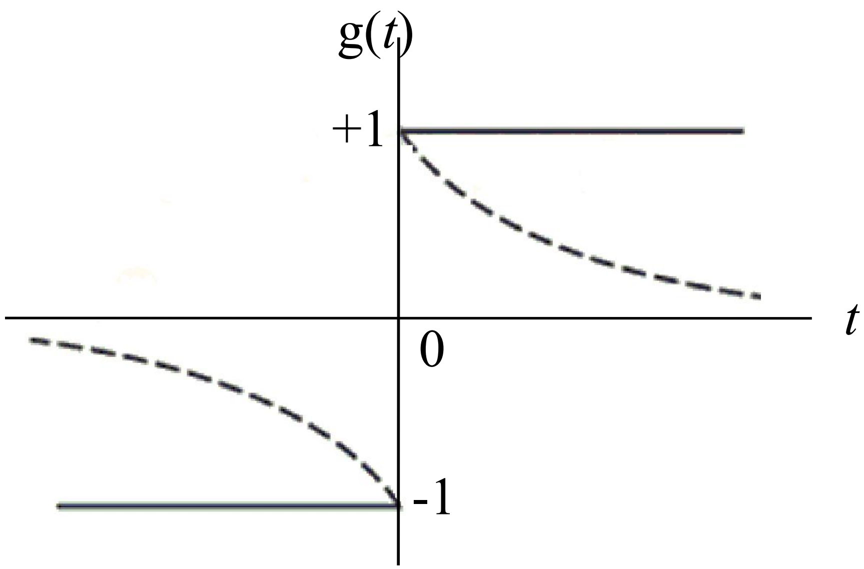
Can be approximated by

$$g(t) = \begin{cases} \exp(-at), & t > 0 \\ 0, & t = 0 \\ -\exp(at), & t < 0 \end{cases}$$

$$g(t) = \exp(-a|t|)\ \mathrm{sgn}(t)$$

# Fourier Transform of Some Useful Functions

Fourier Transform of sgn($t$) function




$$\mathbf{F}[\exp(-a|t|)\,\mathrm{sgn}(t)] = \frac{1}{a + j2\pi f} - \frac{1}{a - j2\pi f} = \frac{-j4\pi f}{a^2 + (2\pi f)^2}$$

$$\mathrm{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

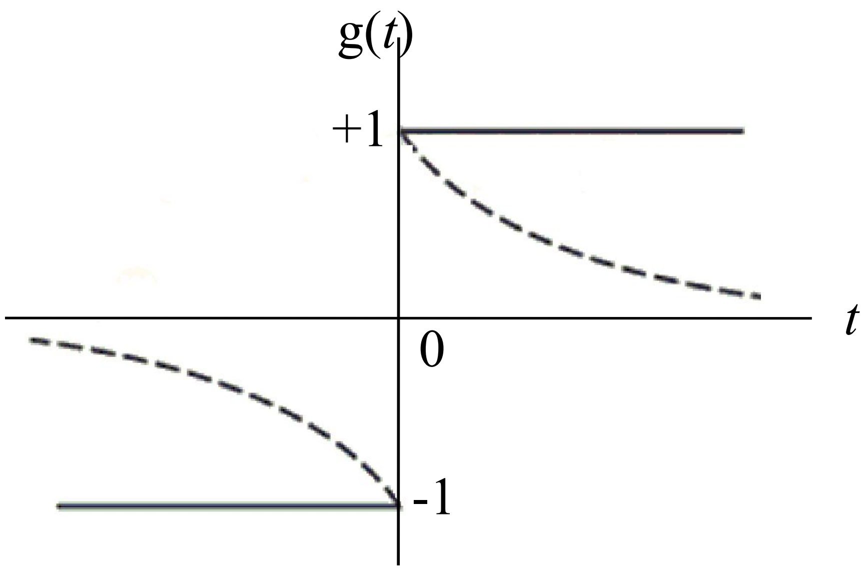
Can be approximated by

$$g(t) = \begin{cases} \exp(-at), & t > 0 \\ 0, & t = 0 \\ -\exp(at), & t < 0 \end{cases}$$

$$g(t) = \exp(-a|t|)\,\mathrm{sgn}(t)$$

# Fourier Transform of Some Useful Functions

## Fourier Transform of sgn(*t*) function




$$F[\exp(-a|t|)\,\mathrm{sgn}(t)] = \frac{1}{a + j2\pi f} - \frac{1}{a - j2\pi f}$$
$$= \frac{-j4\pi f}{a^2 + (2\pi f)^2}$$

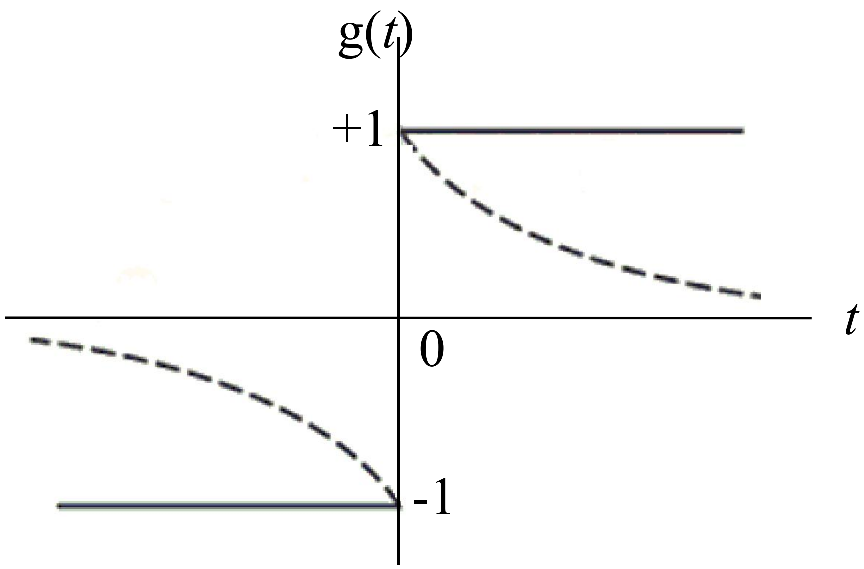
Limiting a to 0

$$F[\mathrm{sgn}(t)] = \lim_{a \to 0} \frac{-4j\pi f}{a^2 + (2\pi f)^2}$$
$$= \frac{1}{j\pi f}$$

$$\mathrm{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

# Fourier Transform of Some Useful Functions

## Fourier Transform of sgn($t$) function




$$\mathrm{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

$$\mathbf{F}[\exp(-a|t|)\,\mathrm{sgn}(t)] = \frac{1}{a + j2\pi f} - \frac{1}{a - j2\pi f} = \frac{-j4\pi f}{a^2 + (2\pi f)^2}$$

Limiting a to 0

$$\mathbf{F}[\mathrm{sgn}(t)] = \lim_{a \to 0} \frac{-4j\pi f}{a^2 + (2\pi f)^2} = \frac{1}{j\pi f}$$

Fourier Transform Pair

$$\mathrm{sgn}(t) \rightleftharpoons \frac{1}{j\pi f}$$